

$\mathcal{H}_\infty$ Control and Estimation — The LMI Solution —

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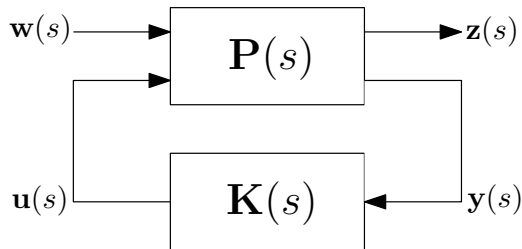
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The Standard Control Problem

- ▶ Consider a **generalized plant**, $\mathbf{P}(s)$, connected in feedback with a **controller**, $\mathbf{K}(s)$.



- ▶ Inputs and outputs are as follows.
 - ▶ $\mathbf{w} \in \mathbb{R}^{n_w}$: exogenous inputs.
 - ▶ $\mathbf{z} \in \mathbb{R}^{n_z}$: regulated output, also called performance output.
 - ▶ $\mathbf{u} \in \mathbb{R}^{n_u}$: control input (controller output [applied by actuators]).
 - ▶ $\mathbf{y} \in \mathbb{R}^{n_y}$: plant output (controller input [a function of measurements]).
- ▶ Design $\mathbf{K}(s)$ to minimize (in terms of some sort of “cost/objective function”) the impact of $\mathbf{w}$ on $\mathbf{z}$.

The Static Full-State $\mathcal{H}_\infty$ Control Design Problem

- In this case, the generalized plant is of the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}_1\mathbf{w}(t) + \mathbf{B}_2\mathbf{u}(t), \quad \mathbf{x}_0 = \mathbf{x}(0), \quad (1)$$

$$\mathbf{z}(t) = \mathbf{C}_1\mathbf{x}(t) + \mathbf{D}_{11}\mathbf{w}(t) + \mathbf{D}_{12}\mathbf{u}(t), \quad (2)$$

$$\mathbf{y}(t) = \mathbf{x}(t) + \mathbf{0}\mathbf{w}(t) + \mathbf{0}\mathbf{u}(t), \quad (3)$$

so that $\mathbf{C}_2 = \mathbf{1}$, $\mathbf{D}_{21} = \mathbf{0}$, and $\mathbf{D}_{22} = \mathbf{0}$.

- In this case, $\mathbf{y}(t) = \mathbf{x}(t)$ (i.e., all states are measured).
- Consider a controller form

$$\begin{aligned} \mathbf{y}_K(t) &= \mathbf{D}_K\mathbf{u}_K(t), \\ \mathbf{u}(t) &= -\mathbf{K}\mathbf{y}(t) \end{aligned}$$

where $\mathbf{y}_K(t) = \mathbf{u}(t)$, $\mathbf{u}_K(t) = \mathbf{y}(t) = \mathbf{x}(t)$, and $\mathbf{D}_K = -\mathbf{K}$.

- This is a static full-state feedback controller.
- In this case, negative feedback is assumed. Could just as well pick $\mathbf{D}_K = \mathbf{K}$ and, in many books/papers, this is what is done.

The Closed-Loop System

- ▶ The closed-loop system is

$$\dot{\mathbf{x}}(t) = (\mathbf{A} - \mathbf{B}_2\mathbf{K})\mathbf{x}(t) + \mathbf{B}_1\mathbf{w}(t), \quad \mathbf{x}_0 = \mathbf{x}(0), \quad (4)$$

$$\mathbf{z}(t) = (\mathbf{C}_1 - \mathbf{D}_{12}\mathbf{K})\mathbf{x}(t) + \mathbf{D}_{11}\mathbf{w}(t), \quad (5)$$

or

$$\dot{\mathbf{x}}(t) = \mathbf{A}_{zw}\mathbf{x}(t) + \mathbf{B}_{zw}\mathbf{w}(t), \quad \mathbf{x}_0 = \mathbf{x}(0),$$

$$\mathbf{z}(t) = \mathbf{C}_{zw}\mathbf{x}(t) + \mathbf{D}_{zw}\mathbf{w}(t),$$

where $\mathbf{A}_{zw} = \mathbf{A} - \mathbf{B}_2\mathbf{K}$, $\mathbf{B}_{zw} = \mathbf{B}_1$, $\mathbf{C}_{zw} = \mathbf{C}_1 - \mathbf{D}_{12}\mathbf{K}$, and $\mathbf{D}_{zw} = \mathbf{D}_{11}$.

- ▶ The associated closed-loop transfer matrix is

$$\begin{aligned} \mathbf{T}_{zw}(s) &= \mathbf{C}_{zw}(s\mathbf{1} - \mathbf{A}_{zw})^{-1}\mathbf{B}_{zw} + \mathbf{D}_{zw} \\ &= (\mathbf{C}_1 - \mathbf{D}_{12}\mathbf{K})(s\mathbf{1} - (\mathbf{A} - \mathbf{B}_2\mathbf{K}))^{-1}\mathbf{B}_1 + \mathbf{D}_{11} \end{aligned}$$

The Optimization Problem

The control design problem at hand is

$$\begin{array}{ll} \underset{\mathbf{K} \in \mathbb{R}^{n_u \times n_x}}{\text{minimize}} & \|\mathcal{T}_{zw}\|_{\infty} \\ \text{such that} & \text{closed-loop system is asymptotically stable,} \end{array}$$

where $\mathbf{z}(t) = (\mathcal{T}_{zw}\mathbf{w})(t)$ and $\mathbf{z}(s) = \mathbf{T}_{zw}(s)\mathbf{w}(s)$.

- Q Why is designing $\mathbf{K} \in \mathbb{R}^{n_u \times n_x}$ to minimize $\|\mathcal{T}_{zw}\|_{\infty}$ a good idea?
- A The impact on $\mathbf{z}$ given a “worst case” $\mathbf{w}$ will be minimized. That is, design $\mathbf{K} \in \mathbb{R}^{n_u \times n_x}$ to minimize the impact of a worst case disturbance.
- The solution that uses convex optimization and LMIs will be considered.

Using Convex Optimization and LMIs to Compute the $\mathcal{H}_\infty$ Norm

Theorem

Consider $\mathbf{G}(s) = \mathbf{C}(s\mathbf{1} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ where $\mathbf{A}$ is Hurwitz, $(\mathbf{A}, \mathbf{B})$ is controllable, and $(\mathbf{A}, \mathbf{C})$ is observable. The following conditions are equivalent.

1. $\|\mathcal{G}\|_\infty < \gamma$.
2. There exists $\mathbf{P} = \mathbf{P}^\top > 0$ such that

$$\mathbf{F}_1 = \begin{bmatrix} \mathbf{PA} + \mathbf{A}^\top \mathbf{P} + \mathbf{C}^\top \mathbf{C} & \mathbf{PB} + \mathbf{C}^\top \mathbf{D} \\ * & \mathbf{D}^\top \mathbf{D} - \gamma^2 \mathbf{1} \end{bmatrix} < 0.$$

3. There exists $\bar{\mathbf{P}} = \bar{\mathbf{P}}^\top > 0$ such that

$$\mathbf{F}_2 = \begin{bmatrix} \bar{\mathbf{P}}\mathbf{A} + \mathbf{A}^\top \bar{\mathbf{P}} & \bar{\mathbf{P}}\mathbf{B} & \mathbf{C}^\top \\ * & -\gamma \mathbf{1} & \mathbf{D}^\top \\ * & * & -\gamma \mathbf{1} \end{bmatrix} < 0.$$

4. There exists $\bar{\mathbf{Q}} = \bar{\mathbf{Q}}^\top > 0$ such that

$$\mathbf{F}_3 = \begin{bmatrix} \mathbf{A}\bar{\mathbf{Q}} + \bar{\mathbf{Q}}\mathbf{A}^\top & \mathbf{B} & \bar{\mathbf{Q}}\mathbf{C}^\top \\ * & -\gamma \mathbf{1} & \mathbf{D}^\top \\ * & * & -\gamma \mathbf{1} \end{bmatrix} < 0.$$

Theorem (Schur Complement Lemma)

Let the symmetric matrix $\mathbf{A}$ be partitioned as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{12}^T & \mathbf{A}_{22} \end{bmatrix}.$$

Then,

$$\begin{aligned} \mathbf{A} < 0 &\Leftrightarrow \mathbf{A}_{11} < 0, \quad \mathbf{A}_{22} - \mathbf{A}_{12}^T \mathbf{A}_{11}^{-1} \mathbf{A}_{12} < 0 \\ &\Leftrightarrow \mathbf{A}_{22} < 0, \quad \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{12}^T < 0 \end{aligned}$$

and

$$\begin{aligned} \mathbf{A} > 0 &\Leftrightarrow \mathbf{A}_{11} > 0, \quad \mathbf{A}_{22} - \mathbf{A}_{12}^T \mathbf{A}_{11}^{-1} \mathbf{A}_{12} > 0 \\ &\Leftrightarrow \mathbf{A}_{22} > 0, \quad \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{12}^T > 0 \end{aligned}$$

Theorem (Congruence Transformation)

Consider $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and $\mathbf{W} \in \mathbb{R}^{n \times n}$ with $\text{rank}(\mathbf{W}) = n$. Then,

$$\mathbf{Q} < 0 \Leftrightarrow \mathbf{W} \mathbf{Q} \mathbf{W}^T < 0.$$

Problem ($\mathcal{H}_\infty$ Control Problem)

Design a static full-state feedback gain $\mathbf{K}$ such that $\|\mathcal{T}_{zw}\|_\infty < \gamma$ holds for $0 < \gamma < \infty$.

Theorem

Given $0 < \gamma < \infty$, the $\mathcal{H}_\infty$ control problem has a solution iff there exists $\mathbf{X} \in \mathbb{S}^{n_x \times n_x}$ and $\mathbf{F} \in \mathbb{R}^{n_u \times n_x}$ such that $\mathbf{X} > 0$,

$$\begin{bmatrix} \mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^\top - \mathbf{B}_2\mathbf{F} - \mathbf{F}^\top\mathbf{B}_2^\top & \mathbf{B}_1 & (\mathbf{C}_1\mathbf{X} - \mathbf{D}_{12}\mathbf{F})^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0. \quad (6)$$

The static full-state feedback gain is given by $\mathbf{K} = \mathbf{F}\mathbf{X}^{-1}$.

- ▶ The design variables are $\mathbf{X}$ and $\mathbf{F}$ (but not γ , which is given).
- ▶ As stated, the above theorem is a *feasibility problem* (i.e., not yet an optimization problem).
- ▶ Reminder: the “*” is just the symmetric part of the matrix in question.

Proof

- Recall that the closed-loop transfer matrix is

$$\begin{aligned}\mathbf{T}_{zw}(s) &= \mathbf{C}_{zw}(s\mathbf{1} - \mathbf{A}_{zw})^{-1}\mathbf{B}_{zw} + \mathbf{D}_{zw} \\ &= (\mathbf{C}_1 - \mathbf{D}_{12}\mathbf{K})(s\mathbf{1} - (\mathbf{A} - \mathbf{B}_2\mathbf{K}))^{-1}\mathbf{B}_1 + \mathbf{D}_{11}.\end{aligned}$$

- Also recall that $\|\mathcal{T}_{zw}\|_{\infty} < \gamma$ iff exists $\mathbf{X} = \mathbf{X}^T > 0$ such that

$$\begin{bmatrix} \mathbf{A}_{zw}\mathbf{X} + \mathbf{X}\mathbf{A}_{zw}^T & \mathbf{B}_{zw} & \mathbf{X}\mathbf{C}_{zw}^T \\ * & -\gamma\mathbf{1} & \mathbf{D}_{zw}^T \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0.$$

Proof Continued

- Expanding the above inequality

$$\begin{aligned} & \begin{bmatrix} (\mathbf{A} - \mathbf{B}_2 \mathbf{K})\mathbf{X} + \mathbf{X}(\mathbf{A} - \mathbf{B}_2 \mathbf{K})^\top & \mathbf{B}_1 & \mathbf{X}(\mathbf{C}_1 - \mathbf{D}_{12} \mathbf{K})^\top \\ * & -\gamma \mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma \mathbf{1} \end{bmatrix} < 0, \\ \Leftrightarrow & \begin{bmatrix} \mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^\top - \mathbf{B}_2 \mathbf{F} - \mathbf{F}^\top \mathbf{B}_2^\top & \mathbf{B}_1 & \mathbf{X}\mathbf{C}_1^\top - \mathbf{F}^\top \mathbf{D}_{12}^\top \\ * & -\gamma \mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma \mathbf{1} \end{bmatrix} < 0, \end{aligned}$$

where $\mathbf{F} = \mathbf{K}\mathbf{X}$.

Optimal $\mathcal{H}_\infty$ Controller

The smallest γ such that $\|\mathcal{T}_{zw}\|_\infty < \gamma$ can be found by solve the following convex optimization problem subject to LMI constraints:

$$\begin{aligned} & \underset{\mathbf{X}=\mathbf{X}^\top, \mathbf{F}, \gamma}{\text{minimize}} && \gamma \\ & \text{such that} && \gamma > 0, \mathbf{X} > 0, \\ & && \begin{bmatrix} \mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^\top - \mathbf{B}_2\mathbf{F} - \mathbf{F}^\top\mathbf{B}_2^\top & \mathbf{B}_1 & (\mathbf{C}_1\mathbf{X} - \mathbf{D}_{12}\mathbf{F})^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0. \end{aligned}$$

The optimal static state-feedback gain (in an $\mathcal{H}_\infty$ sense) is then $\mathbf{K} = \mathbf{F}\mathbf{X}^{-1}$.

- ▶ The above is an *optimization problem*. The design (optimization) variables are $\mathbf{X}$, $\mathbf{F}$, and γ .
- ▶ Note, the word “optimal” is meaningless without specifying “with respect to what” (i.e., the form of the cost function). In this case the $\mathbf{K}$ is optimal in an $\mathcal{H}_\infty$ sense.

The $\mathcal{H}_\infty$ Estimation Problem

- Consider

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}_1\mathbf{w}(t) + \mathbf{B}_2\mathbf{u}(t), & \mathbf{x}_0 &= \mathbf{x}(0), \\ \mathbf{y}(t) &= \mathbf{C}_2\mathbf{x}(t) + \mathbf{D}_{21}\mathbf{w}(t)\end{aligned}$$

where $\mathbf{w}$ can be decomposed into process noise (i.e., a disturbance) and measurement noise.

- Consider an observer of the form

$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}_2\mathbf{u}(t) + \mathbf{L}(\mathbf{y}(t) - \hat{\mathbf{y}}(t)) \\ &= \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}_2\mathbf{u}(t) + \mathbf{L}(\mathbf{C}_2\mathbf{x}(t) + \mathbf{D}_{21}\mathbf{w}(t) - \mathbf{C}_2\hat{\mathbf{x}}(t)) \\ &= (\mathbf{A} - \mathbf{L}\mathbf{C}_2)\hat{\mathbf{x}}(t) + \mathbf{B}_2\mathbf{u}(t) + \mathbf{L}\mathbf{C}_2\mathbf{x}(t) + \mathbf{L}\mathbf{D}_{21}\mathbf{w}(t).\end{aligned}$$

- Next, consider the error $\mathbf{e}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t)$, and the error dynamics

$$\begin{aligned}\dot{\mathbf{e}}(t) &= \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}(t) \\ &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}_1\mathbf{w}(t) + \mathbf{B}_2\mathbf{u}(t) - (\mathbf{A} - \mathbf{L}\mathbf{C}_2)\hat{\mathbf{x}}(t) - \mathbf{B}_2\mathbf{u}(t) - \mathbf{L}\mathbf{C}_2\mathbf{x}(t) - \mathbf{L}\mathbf{D}_{21}\mathbf{w}(t) \\ &= (\mathbf{A} - \mathbf{L}\mathbf{C}_2)\mathbf{e}(t) + (\mathbf{B}_1 - \mathbf{L}\mathbf{D}_{21})\mathbf{w}(t)\end{aligned}$$

- Let the performance be $\mathbf{z}(t) = \mathbf{C}_1\mathbf{e}(t) + \mathbf{D}_{11}\mathbf{w}(t)$.

- The transfer matrix $\mathbf{T}_{zw}(s)$ is

$$\begin{aligned}\mathbf{T}_{zw}(s) &= \mathbf{C}_{zw}(s\mathbf{1} - \mathbf{A}_{zw})^{-1}\mathbf{B}_{zw} + \mathbf{D}_{zw} \\ &= \mathbf{C}_1(s\mathbf{1} - (\mathbf{A} - \mathbf{L}\mathbf{C}_2))^{-1}(\mathbf{B}_1 - \mathbf{L}\mathbf{D}_{21}) + \mathbf{D}_{11}.\end{aligned}$$

Problem ($\mathcal{H}_\infty$ Estimation Problem)

Design an observer gain $\mathbf{L}$ such that $\|\mathcal{T}_{zw}\|_\infty < \gamma$ holds for $0 < \gamma < \infty$.

Theorem

Given $0 < \gamma < \infty$, the $\mathcal{H}_\infty$ estimation problem has a solution iff there exists $\mathbf{X} \in \mathbb{S}^{n_x \times n_x}$ and $\mathbf{F} \in \mathbb{R}^{n_x \times n_y}$ such that $\mathbf{X} > 0$

$$\begin{bmatrix} \mathbf{XA} + \mathbf{A}^\top \mathbf{X} - \mathbf{FC}_2 - \mathbf{C}_2^\top \mathbf{F}^\top & \mathbf{XB}_1 - \mathbf{FD}_{21} & \mathbf{C}_1^\top \\ * & -\gamma \mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma \mathbf{1} \end{bmatrix} < 0. \quad (7)$$

The observer gain is given by $\mathbf{L} = \mathbf{X}^{-1}\mathbf{F}$.

- ▶ The design variables are $\mathbf{X}$ and $\mathbf{F}$ (but not γ , which is given).
- ▶ As stated, the above theorem is a *feasibility problem* (i.e., not yet an optimization problem).

Proof

- Recall that the closed-loop transfer matrix is

$$\mathbf{T}_{zw}(s) = \mathbf{C}_{zw}(s\mathbf{1} - \mathbf{A}_{zw})^{-1}\mathbf{B}_{zw} + \mathbf{D}_{zw} = \mathbf{C}_1(s\mathbf{1} - (\mathbf{A} - \mathbf{L}\mathbf{C}_2))^{-1}(\mathbf{B}_1 - \mathbf{L}\mathbf{D}_{21}) + \mathbf{D}_{11}.$$

- Also recall that $\|\mathcal{T}_{zw}\|_\infty < \gamma$ iff exists $\mathbf{X} = \mathbf{X}^\top > 0$ such that

$$\begin{bmatrix} \mathbf{X}\mathbf{A}_{zw} + \mathbf{A}_{zw}^\top\mathbf{X} & \mathbf{X}\mathbf{B}_{zw} & \mathbf{C}_{zw}^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{zw}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0.$$

- Expanding the above inequality

$$\begin{aligned} & \begin{bmatrix} \mathbf{X}(\mathbf{A} - \mathbf{L}\mathbf{C}_2) + (\mathbf{A} - \mathbf{L}\mathbf{C}_2)^\top\mathbf{X} & \mathbf{X}(\mathbf{B}_1 - \mathbf{L}\mathbf{D}_{21}) & \mathbf{C}_1^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0, \\ \Leftrightarrow & \begin{bmatrix} \mathbf{X}\mathbf{A} + \mathbf{A}^\top\mathbf{X} - \mathbf{F}\mathbf{C}_2 - \mathbf{C}_2^\top\mathbf{F}^\top & \mathbf{X}\mathbf{B}_1 - \mathbf{F}\mathbf{D}_{21} & \mathbf{C}_1^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0, \end{aligned}$$

where $\mathbf{F} = \mathbf{X}\mathbf{L}$.

Optimal $\mathcal{H}_\infty$ Estimator

The smallest γ such that $\|\mathcal{T}_{zw}\|_\infty < \gamma$ can be found by solve the following convex optimization problem subject to LMI constraints:

$$\begin{aligned} & \underset{\mathbf{X}=\mathbf{X}^\top, \mathbf{F}, \gamma}{\text{minimize}} && \gamma \\ & \text{such that} && \gamma > 0, \mathbf{X} > 0, \\ & && \begin{bmatrix} \mathbf{X}\mathbf{A} + \mathbf{A}^\top\mathbf{X} - \mathbf{F}\mathbf{C}_2 - \mathbf{C}_2^\top\mathbf{F}^\top & \mathbf{X}\mathbf{B}_1 - \mathbf{F}\mathbf{D}_{21} & \mathbf{C}_1^\top \\ * & -\gamma\mathbf{1} & \mathbf{D}_{11}^\top \\ * & * & -\gamma\mathbf{1} \end{bmatrix} < 0. \end{aligned}$$

The optimal observer gain is then $\mathbf{L} = \mathbf{X}^{-1}\mathbf{F}$.

The above is an *optimization problem*. The design (optimization) variables are $\mathbf{X}$, $\mathbf{F}$, and γ .

References

These slides are based on [1].

- [1] G.-R. Duan and H.-H. Yu, *LMIs in Control Systems: Analysis, Design and Applications*. Boca Raton, FL: Routledge, Taylor and Francis Group, CRC Press, 2013.